# Author result prop-approx (printed 5.3) of ahl-163-stochastic-nonlinear-wave-dynamics-on-compact-2023

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- The statement and its own proof as printed in the article. source lines 2172-2369. the counted unit and the argument carried out in the span

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**Proposition 0.1.** *Let $m, j \in \mathbb{N}$, $N_p \to \infty$ and $(u_0, u_1, \omega) \in \bigcap_{p\in\mathbb{N}} \Sigma^{m,j}_{N_p}$. Then $\{\Phi^{N_p}(t)(u_0, u_1, \omega) - \Psi(t)(u_0, u_1, \omega)\}_{p\in\mathbb{N}}$ is a Cauchy sequence in $C([0, 2^j]; \mathcal{H}^{s_2}(\mathcal{M}))$. In particular, $\{\Phi^{N_p}(t)(u_0, u_1, \omega)\}_{p\in\mathbb{N}}$ is a Cauchy sequence in $C([0, 2^j]; \mathcal{H}^{s}(\mathcal{M}))$. Here $s < 0 < s_2 < s_1 < 1 + s$ with $0 < -s < 1 - s_1 < 1 - s_2 \ll 1$.*

*Proof of Proposition 0.1.* Let us fix $m, j \in \mathbb{N}$, $\delta > 0$ as in (5.9), and $(u_0, u_1, \omega) \in \bigcap_{p\in\mathbb{N}} \Sigma^{m,j}_{N_p}$. In the following, we fix two (large) integers $N, M \in \{N_p\}_{p\in\mathbb{N}}$. Again, we write

$$(w_N, \partial_t w_N)(t) = \Phi^N(t)(u_0, u_1, \omega) - \Psi(t, u_0, u_1, \omega),$$

and we denote by $\Phi^N_1(t)$ (respectively $\Phi^N_2(t)$) the first (respectively second) component of $\Phi^N(t)$. We will control inductively the difference $(w_N(t), \partial_t w_N(t)) - (w_M(t), \partial_t w_M(t))$ on the time intervals $[a\delta, (a+1)\delta]$, $a = 1, \dots, [2^j/\delta]$. We begin by controlling the difference on the first time interval, corresponding to $a = 0$.

Then on $[0, \delta]$, we have for $N \le M$:

$$\begin{aligned}
&w_M(t) - w_N(t)\\
&= \sum_{\ell=0}^{k} \binom{k}{\ell} \int_0^t V(t-t') \Big\{ \mathbf{P}_M \Big[ H_\ell\big(\mathbf{P}_M \Psi(t', u_0, u_1, \omega); \sigma_M(x)\big) \big(\mathbf{P}_M w_M(t')\big)^{k-\ell} \Big]\\
&\quad - \mathbf{P}_N \Big[ H_\ell\big(\mathbf{P}_N \Psi(t', u_0, u_1, \omega); \sigma_N(x)\big) \big(\mathbf{P}_N w_N(t')\big)^{k-\ell} \Big] \Big\} dt'\\
&= \sum_{\ell=0}^{k} \binom{k}{\ell} \int_0^t V(t-t') \Big\{ (\mathbf{P}_M - \mathbf{P}_N) \Big[ H_\ell\Big(\mathbf{P}_M \Psi(t', u_0, u_1, \omega); \sigma_M(x)\Big) \big(\mathbf{P}_M w_M(t')\big)^{k-\ell} \Big]\\
&\quad + \mathbf{P}_N \Big[ \Big( H_\ell\big(\mathbf{P}_M \Psi(t', u_0, u_1, \omega); \sigma_M(x)\big)\\
&\quad - H_\ell\big(\mathbf{P}_N \Psi(t', u_0, u_1, \omega); \sigma_N(x)\big)\Big) \big(\mathbf{P}_M w_M(t')\big)^{k-\ell} \Big]\\
&\quad + \mathbf{P}_N \Big[ H_\ell\big(\mathbf{P}_N \Psi(t', u_0, u_1, \omega); \sigma_N(x)\big) \Big( \big(\mathbf{P}_M w_M(t')\big)^{k-\ell} - \big(\mathbf{P}_N w_N(t')\big)^{k-\ell} \Big) \Big] \Big\} dt'\\
&=: \mathrm{I} + \mathrm{II} + \mathrm{III}.
\end{aligned} \tag{1}$$

To estimate these terms, we proceed as in the proof of Proposition 4.1. We begin with

$$\begin{aligned}
&\| \mathrm{I} \|_{C([0,\delta]; H^{s_2})}\\
&\lesssim \sum_{\ell=0}^{k} \Big\| (\mathbf{P}_M - \mathbf{P}_N) \Big[ H_\ell\Big(\mathbf{P}_M \Psi(u_0, u_1, \omega); \sigma_M(x)\Big) \big(\mathbf{P}_M w_M\big)^{k-\ell} \Big] \Big\|_{L^1([0,\delta]; H^{s_2-1})}\\
&\lesssim \sum_{\ell=0}^{k} N^{s_2-s_1} \Big\| \Big[ H_\ell\Big(\mathbf{P}_M \Psi(u_0, u_1, \omega); \sigma_M(x)\Big) \big(\mathbf{P}_M w_M\big)^{k-\ell} \Big] \Big\|_{L^1([0,\delta]; H^{s_1-1})}\\
&\lesssim \sum_{\ell=0}^{k} N^{s_2-s_1} \delta^{\frac{1}{2}} \Big\| H_\ell\Big(\mathbf{P}_M \Psi(u_0, u_1, \omega); \sigma_M(x)\Big) \Big\|_{L^2([0,1]; W^{s,\infty})} \|\mathbf{P}_M w_M\|^{k-\ell}_{C([0,\delta]; H^{s_1})}\\
&\le C(m, j) N^{s_2-s_1},
\end{aligned}$$

for some constant $C(m, j)$ independent of $N, M$, where the second to last estimate comes from the same argument as in the proof of Proposition 4.1, and the last one from the condition (5.10) given by $(u_0, u_1, \omega) \in \mathcal{B}^{m,j}_M(D)$, with (5.11) and the choice of $\delta$ in (5.9).

Similarly, we bound

$$
\begin{aligned}
&\|\mathrm{II}\|_{C([0,\delta];H^{s_2})}\\
&\lesssim \sum_{\ell=1}^{k} \Big\| \Big[ H_\ell\big(\mathbf{P}_M \Psi(u_0,u_1,\omega);\sigma_M(x)\big)\\
&\qquad - H_\ell\big(\mathbf{P}_N \Psi(u_0,u_1,\omega);\sigma_N(x)\big)\Big] \big(\mathbf{P}_M w_M\big)^{k-\ell}\Big\|_{L^1([0,\delta];H^{s_2-1})}\\
&\lesssim \delta^{\frac12}\Big\{ \Big\| H_k\big(\mathbf{P}_M \Psi(u_0,u_1,\omega);\sigma_M(x)\big) - H_k\big(\mathbf{P}_N \Psi(u_0,u_1,\omega);\sigma_N(x)\big)\Big\|_{L^2([0,\delta];H^{s_2-1})}\\
&\qquad + \sum_{\ell=1}^{k-1} \Big\| H_\ell\big(\mathbf{P}_M \Psi(u_0,u_1,\omega);\sigma_M(x)\big) - H_\ell\big(\mathbf{P}_N \Psi(u_0,u_1,\omega);\sigma_N(x)\big)\Big\|_{L^2([0,\delta];W^{s,\infty})}\\
&\qquad\qquad \times \big\|\mathbf{P}_M w_M\big\|^{k-\ell}_{C([0,\delta];H^{s_1})}\Big\}\\
&\le C(m,j) N^{-\varepsilon},
\end{aligned}
$$

where the last bound comes from (5.10) given by $(u_0,u_1,\omega)\in \mathcal{B}_M^{m,j}(D)$ and from (5.11) with the choice of $\delta$.

Finally, we can further decompose

$$
\begin{aligned}
\mathrm{III} = \sum_{\ell=0}^{k-1}\binom{k}{\ell}\int_0^t V(t-t')\mathbf{P}_N \Big\{ & H_\ell\big(\mathbf{P}_N\Psi(t',u_0,u_1,\omega);\sigma_N(x)\big)\\
&\Big[\Big(w_N(t')^{k-\ell} - \big(\mathbf{P}_N w_N(t')\big)^{k-\ell}\Big) + \Big(\big(\mathbf{P}_M w_M(t')\big)^{k-\ell} - w_M(t')^{k-\ell}\Big)\\
&+ \Big(w_M(t')^{k-\ell} - w_N(t')^{k-\ell}\Big)\Big]\Big\} dt' =: \mathrm{III}_1 + \mathrm{III}_2 + \mathrm{III}_3.
\end{aligned}
$$

We estimate similarly as before

$$
\begin{aligned}
\|\mathrm{III}_1\|_{C([0,\delta];H^{s_2})} &\lesssim \sum_{\ell=0}^{k-1} \delta^{\frac12}\big\| H_\ell\big(\mathbf{P}_N\Psi(u_0,u_1,\omega);\sigma_N(x)\big)\big\|_{L^2([0,1];W^{s,\infty})}\\
&\qquad \times \big\|(1-\mathbf{P}_N)w_N\big\|_{C([0,\delta];H^{s_2})}\Big(\big\|\mathbf{P}_N w_N\big\|^{k-\ell-1}_{C([0,\delta];H^{s_2})} + \big\|w_N\big\|^{k-\ell-1}_{C([0,\delta];H^{s_2})}\Big)\\
&\le C(m,j) N^{s_2-s_1},
\end{aligned}
$$

where the first estimate follows from the same argument as in the proof of Proposition 4.1, provided that $s_2 < s_1 < 1+s < 1$ is close enough to 1. The same argument applies to $\mathrm{III}_2$ and gives the same bound (with $M$ in place of $N$), and the last term can be bounded similarly by

$$
\begin{aligned}
\big\|\mathrm{III}_3\big\|_{C([0,\delta];H^{s_2})} &\lesssim \sum_{\ell=0}^{k-1} \delta^{\frac12}\big\| H_\ell\big(\mathbf{P}_N\Psi(u_0,u_1,\omega);\sigma_N(x)\big)\big\|_{L^2([0,1];W^{s,\infty})}\\
&\qquad \times \big\|w_M - w_N\big\|_{C([0,\delta];H^{s_2})}\Big(\big\|w_M\big\|^{k-\ell-1}_{C([0,\delta];H^{s_2})} + \big\|w_N\big\|^{k-\ell-1}_{C([0,\delta];H^{s_2})}\Big)\\
&\le C\delta^{\frac12} D(m+j)^{\frac{k}{2}} \|w_M - w_N\|_{C([0,\delta];H^{s_2})},
\end{aligned}
$$

where the last estimate comes again from (5.10) thanks to $(u_0,u_1,\omega)\in\mathcal{B}_M^{m,j}(D)$, and from (5.11) with the argument of Proposition 4.1 applied with $s_2$ (provided

that $s_2$ is sufficiently close to 1). With our choice of $C\delta^{\frac{1}{2}} D(m+j)^{\frac{k}{2}} = C\delta^{\frac{1}{2}} R \leq \frac{1}{2}$, we can absorb this last term in the left-hand side of (1).

The same arguments also apply to control $\Phi_2^N(t) - \Phi_2^M(t)$ on $[0,\delta]$. Therefore, gathering the estimates above leads to

$$\left\|\Phi^M(t)(u_0,u_1,\omega) - \Phi^N(t)(u_0,u_1,\omega)\right\|_{C([0,\delta];\mathcal{H}^{s_2})} \leq C(m,j)\big(N^{s_2-s_1} + N^{-\varepsilon}\big) \quad (2)$$

for any $N \leq M$. In particular this shows the convergence on the time interval $[0,\delta]$.

We now investigate the convergence on the second time interval $[\delta, 2\delta]$: we first decompose

$$\begin{aligned}\left\|\Phi^M(t+\delta)(u_0,u_1,\omega) - \Phi^N(t+\delta)(u_0,u_1,\omega)\right\|_{C([0,\delta];\mathcal{H}^{s_2})} \\ \leq \left\|\Phi^M(t)\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega) - \Phi^N(t)\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\right\|_{C([0,\delta];\mathcal{H}^{s_2})} \\ + \left\|\Phi^N(t)\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega) - \Phi^N(t)\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right\|_{C([0,\delta];\mathcal{H}^{s_2})}.\end{aligned}$$

Note that replacing $\Psi(u_0,u_1,\omega)$ by $\Psi(\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega))$ in the previous estimates and using that we still have $\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega) \in \mathcal{B}_M^{m,j}$ by choice of $(u_0,u_1,\omega) \in \Sigma_M^{m,j}$ shows that the first term above is still bounded by

$$\begin{aligned}\left\|\Phi^M(t)\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega) - \Phi^N(t)\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\right\|_{C([0,\delta];\mathcal{H}^{s_2})} \\ \leq C(m,j)\left(N^{s_2-s_1} + N^{-\varepsilon}\right)\end{aligned}$$

for any $N \leq M$.

Thus we need to deal with the second term. We can redefine

$$w_M(t) = \Phi_1^N(t)\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega) - \Psi\left(t, \widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\right)$$

and

$$w_N(t) = \Phi_1^N(t)\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega) - \Psi\left(t, \widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right),$$

and since $N, M \in \{N_p\}_{p\in\mathbb{N}}$ and $(u_0,u_1,\omega) \in \bigcap_{p\in\mathbb{N}} \Sigma_{N_p}^{m,j}$ we have in particular that both $\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega) \in \mathcal{B}_M^{m,j}(D)$ and $\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega) \in \mathcal{B}_N^{m,j}(D)$ so that both $w_M$ and $w_N$ are well-defined and enjoy the bound (5.11). Moreover, by definition of $\Phi_1^N(t)$ and $\Psi$, they satisfy the following Duhamel formula:

$$\begin{aligned}w_M = \sum_{\ell=0}^{k} \binom{k}{\ell} \int_0^t V(t-t')\mathbf{P}_N \\ \left\{H_\ell\Big(\mathbf{P}_N\Psi\big(t', \widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\big); \sigma_N(x)\Big)\big(\mathbf{P}_N w_M(t')\big)^{k-\ell}\right\}dt',\end{aligned}$$

and

$$\begin{aligned}w_N = \sum_{\ell=0}^{k} \binom{k}{\ell} \int_0^t V(t-t')\mathbf{P}_N \\ \left\{H_\ell\Big(\mathbf{P}_N\Psi\big(t', \widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\big); \sigma_N(x)\Big)\big(\mathbf{P}_N w_N(t')\big)^{k-\ell}\right\}dt'.\end{aligned}$$

To estimate in $C([0,\delta];H^{s_2}(\mathcal{M}))$ the difference

$$\begin{aligned}\Phi_1^N(t)\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)-\Phi_1^N(t)\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\\ =\Psi\big(t,\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\big)-\Psi\big(t,\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\big)+w_M-w_N,\end{aligned}$$

we first bound directly the linear terms by

$$\begin{aligned}\Big\|\Psi\Big(t,\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\Big)-\Psi\Big(t,\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\Big)\Big\|_{C([0,\delta];H^{s_2})}\\ \le\big\|\Phi^M(\delta)(u_0,u_1,\omega)-\Phi^N(\delta)(u_0,u_1,\omega)\big\|_{\mathcal{H}^{s_2}}\\ \le C(m,j)\left(N^{s_2-s_1}+N^{-\varepsilon}\right)\end{aligned}$$

thanks to (1.11) and (2).

To estimate the difference of the nonlinear components, we decompose

$$\begin{aligned}&w_M-w_N\\ &=\sum_{\ell=0}^{k}\binom{k}{\ell}\int_0^t V(t-t')\mathbf{P}_N\Big\{H_\ell\Big(\mathbf{P}_N\Psi\left(t',\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\\ &\quad\big(\mathbf{P}_Nw_M(t')\big)^{k-\ell}-H_\ell\Big(\mathbf{P}_N\Psi\left(t',\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\big(\mathbf{P}_Nw_N(t')\big)^{k-\ell}\Big\}dt'\\ &=\sum_{\ell=0}^{k}\binom{k}{\ell}\int_0^t V(t-t')\mathbf{P}_N\Big\{\Big[H_\ell\Big(\mathbf{P}_N\Psi\big(t',\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\big);\sigma_N(x)\Big)\\ &\quad-H_\ell\Big(\mathbf{P}_N\Psi\left(t',\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\Big]\big(\mathbf{P}_Nw_M(t')\big)^{k-\ell}\\ &\quad+H_\ell\Big(\mathbf{P}_N\Psi\left(t',\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\\ &\qquad\qquad\Big[\big(\mathbf{P}_Nw_M(t')\big)^{k-\ell}-\big(\mathbf{P}_Nw_N(t')\big)^{k-\ell}\Big]\Big\}dt'\\ &=:\widetilde{\mathbb{II}}+\widetilde{\mathbb{III}}.\end{aligned}$$

First note that we can estimate $\widetilde{\mathbb{III}}$ exactly as $\mathbb{III}$ in (1), giving the bound

$$\big\|\widetilde{\mathbb{III}}\big\|_{C([0,\delta];H^{s_2})}\le C(m,j)N^{s_2-s_1}+\frac{1}{2}\big\|w_M-w_N\big\|_{C([0,\delta];H^{s_2})}.$$

Finally, we estimate the remaining term by

$$\begin{aligned}\big\|\widetilde{\mathbb{II}}\big\|_{C([0,\delta];H^{s_2})}\lesssim\sum_{\ell=1}^{k}\Big\|\Big[H_\ell\Big(\mathbf{P}_N\Psi\left(\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\\ -H_\ell\Big(\mathbf{P}_N\Psi\left(\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\Big]\big(\mathbf{P}_Nw_M\big)^{k-\ell}\Big\|_{L^1([0,\delta];H^{s_2-1})}\end{aligned}$$

Writing then

$$H_\ell(u;\sigma_N)-H_\ell(v;\sigma_N)=-\sum_{i=0}^{\ell-1}\binom{\ell}{i}H_i(u;\sigma_N)(v-u)^{\ell-i}$$

thanks to (3.1), we can then estimate the previous term with

$$\begin{aligned}
&\sum_{\ell=1}^{k}\sum_{i=0}^{\ell-1}\delta^{\frac{1}{2}}\Big\|\Big[\mathbf{P}_N\Psi\left(\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\right)-\mathbf{P}_N\Psi\left(\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right)\Big]^{\ell-i}\\
&\times H_i\Big(\mathbf{P}_N\Psi\left(\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\big(\mathbf{P}_N w_M\big)^{k-\ell}\Big\|_{L^2([0,\delta];H^{s_2-1})}\\
\lesssim&\sum_{\ell=1}^{k}\sum_{i=0}^{\ell-1}\delta^{\frac{1}{2}}\Big\|H_i\Big(\mathbf{P}_N\Psi\left(\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right);\sigma_N(x)\Big)\Big\|_{L^2([0,1];W^{s,\infty})}\\
&\times\Big\|\Psi\left(\widetilde{\Phi}^M(\delta)(u_0,u_1,\omega)\right)-\Psi\left(\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\right)\Big\|^{\ell-i}_{C([0,\delta];H^{s_2})}\big\|w_M\big\|^{k-\ell}_{C([0,\delta];H^{s_2})},
\end{aligned}$$

provided again that $s_2$ is close enough to 1 (depending on $k$). Using then that $\widetilde{\Phi}^N(\delta)(u_0,u_1,\omega)\in\mathcal{B}_N^{m,j}(D)$ and (5.11),(2), we finally get

$$\big\|\widetilde{\mathbb{I}}\big\|_{C([0,\delta];H^{s_2})}\leq C(m,j)\left(N^{s_2-s_1}+N^{-\varepsilon}\right).$$

Gathering the estimates above, we obtain

$$\big\|w_M-w_N\big\|_{C([0,\delta];H^{s_2})}\leq 4C(m,j)\left(N^{s_2-s_1}+N^{-\varepsilon}\right)$$

which leads to

$$\big\|(w_N,\partial_t w_N)(t+\delta)-(w_M,\partial_t w_M)(t+\delta)\big\|_{C([0,\delta];\mathcal{H}^{s_2})}\leq C_2(m,j)\left(N^{s_2-s_1}+N^{-\varepsilon}\right)$$

for some larger constant $C_2(m,j)\geq C(m,j)$. This shows that $\{(w_{N_p},\partial_t w_{N_p})\}_{p\in\mathbb{N}}$ is also a Cauchy sequence in $C([0,2\delta];\mathcal{H}^{s_2}(\mathcal{M}))$.

We can then proceed inductively on $a=0,\dots,[2^j/\delta]$ and repeat the previous estimates by using that at each step $\widetilde{\Phi}^N(a\delta)(u_0,u_1,\omega)\in\mathcal{B}_N^{m,j}(D)$ since $(u_0,u_1,\omega)\in\Sigma_N^{m,j}$. Thus we deduce that there exists a (large) constant $C_{2^j/\delta}(m,j)>0$ such that for any $N,M\in\{N_p\}_{p\in\mathbb{N}}$ with $N\leq M$ it holds

$$\big\|(w_N,\partial_t w_N)-(w_M,\partial_t w_M)\big\|_{C([0,2^j];\mathcal{H}^{s_2})}\leq C_{2^j/\delta}(m,j)\left(N^{s_2-s_1}+N^{-\varepsilon}\right). \quad (3)$$

This is enough to show the convergence of $\{\Phi^{N_p}(t)(u_0,u_1,\omega)-\Psi(t)(u_0,u_1,\omega)\}$ in $C([0,2^j];\mathcal{H}^{s_2}(\mathcal{M}))$. As a result, $\left\{\Phi^{N_p}(t)(u_0,u_1,\omega)\right\}_{p\in\mathbb{N}}$ converges in $C([0,2^j];\mathcal{H}^s(\mathcal{M}))$. This concludes the proof of Proposition 0.1. □